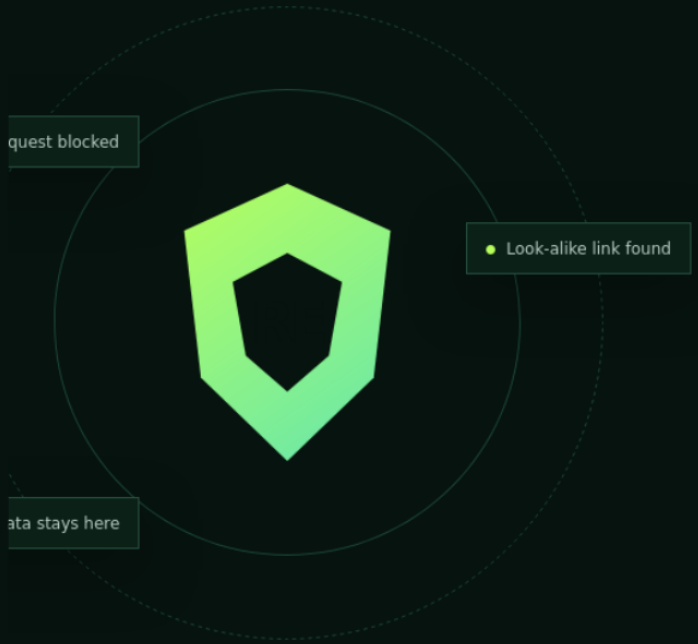
ON-DEVICE SOCIAL-ENGINEERING FIREWALL

Rakshak Edge

A private safety check before the user clicks, pays, or shares a secret



SNAPDRAGON X SERIES

Hariharan C | 2026

The critical moment happens before the tap

A suspicious message exploits urgency and trust. Sending it to a cloud checker can expose the same private content the user is trying to protect.

01

Private content

OTP messages, account alerts, payment handles, and personal context.

02

Fast manipulation

Threats and rewards push the user to act before checking another channel.

03

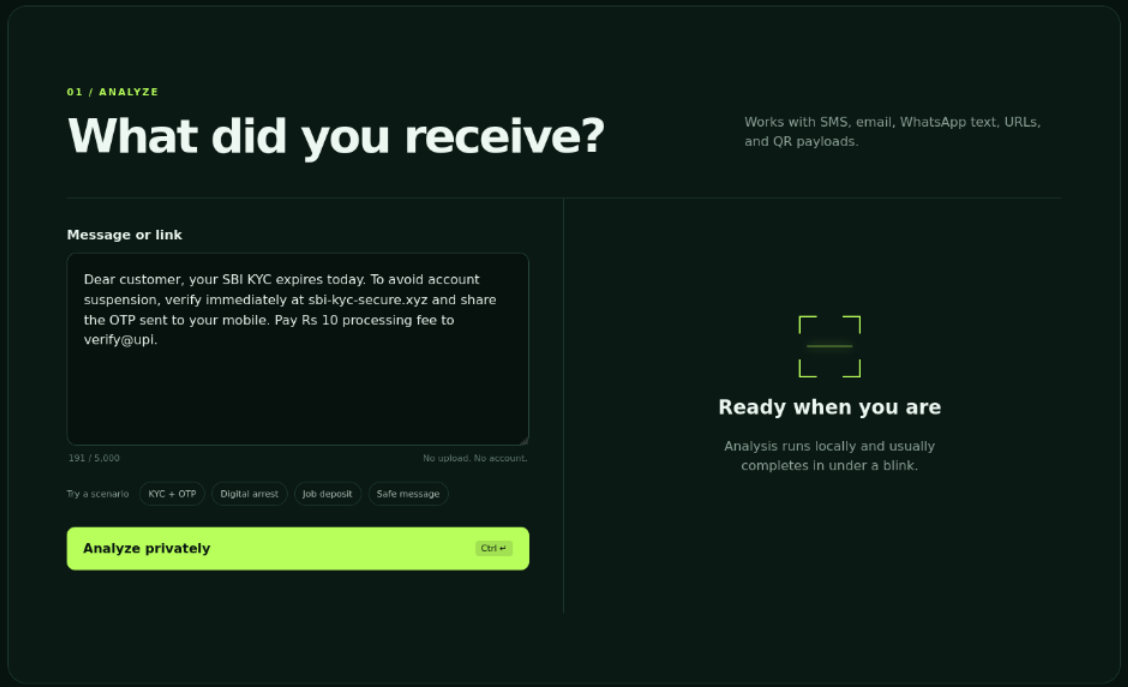
Opaque verdicts

A score alone cannot teach the user what triggered the warning or what to do next.

Design question

Can a PC explain the risk locally, in the moment that matters?

One local workflow from suspicion to safe action



Paste

Message, URL, UPI request, or QR payload

Understand

Risk score with visible evidence and extracted entities

Act safely

Independent verification path without another risky link

Scams unfold across a conversation

Rakshak connects the stages that single-message classifiers lose.

01	IDENTITY SETUP	Hi, this is Arun from finance. Save my new number.	42
02	ISOLATION	I am boarding a flight. Do not call. Keep this confidential.	55
03	EXTRACTION	Purchase five vouchers and send the codes before 4 PM.	72

81 /100

COORDINATED SCAM

Identity setup
Isolation and pressure
Money or access request

The raw conversation is processed in memory and never appears in the exported report.

Hybrid reasoning keeps security explainable

Transparent signals stay authoritative. A compact classifier adds context today; the production profile moves the encoder to the Hexagon NPU.

01

Input boundary

Loopback API
No persistence

>

02

Evidence engines

URL + entity rules
Local semantic model

>

03

Snapdragon profile

INT8 encoder
ONNX Runtime + QNN

>

04

Decision layer

Calibrated score
Reasons + safe action

Graceful fallback

QNN unavailable? ONNX Runtime uses CPU. Model absent? The deterministic engine still delivers a complete, testable product.

AI catches the scam without a malicious link

72 /100

HIGH RISK

Stop and verify independently

Semantic scam pattern	99.8% high-risk confidence	AI
Voucher purchase	Financial coercion	RULE
No URL	Nothing for a link scanner	0
No OTP or UPI	Meaning comes from context	0

SAFEST NEXT STEP

Call the finance head using the number already in your contacts.

A locked suite tests the complete product

97.78%

45-case system suite

15 / 15

high-risk cases

15 / 15

medium-risk cases

10 / 10

regression tests

ONE DISCLOSED ERROR

A safety reminder about OTP and PIN is conservatively rated medium instead of low.

39

The system suite is balanced, never used for training, and prints every error. The model-only holdout remains 94.44%.

Local protection is the product advantage

0

cloud calls by default

< 1 ms

median reference CPU

INT8

production model target

45 TOPS

Snapdragon X Elite NPU

Same private API, stronger production encoder

The reference model makes the complete demo reproducible today. On Snapdragon X Series, the production profile swaps in an INT8 encoder through ONNX Runtime QNN while keeping the evidence layer, conversation logic, and offline boundary unchanged.

A safety product needs honest failure behavior

01	**Runs today**	One command, standard library server, offline message and thread APIs
02	**Private**	No raw history; exported reports exclude the source conversation
03	**Accessible**	Keyboard flow, high contrast, reduced motion, plain-language actions
04	**Deployable**	QNN adapter, INT8 model contract, CPU fallback, acceptance checks
05	**Bounded**	Risk indicator, not a fraud verdict; one false positive disclosed

```
python app.py → localhost:8765
```

RAKSHAK EDGE

A safer pause before the tap

Conversation-aware protection that stays on the PC

LIVE DEMO

```
python app.py
```

Working demo, locked evaluation, privacy threat model, Snapdragon profile

Hariharan C